

Interpretation

Table: inventory

Purpose:

Stores all product details available for ordering — each record represents one item (e.g., milk tea flavor, add-on, packaging).

Key Columns:

- `id` (*PK*) – Unique identifier for each product.
- `name` – Item name (e.g., “Wintermelon Milk Tea”).
- `description` – Optional product details.
- `price` – Standard selling price per unit.
- `stock_qty` – Current stock quantity available.
- `is_active` – Indicates if the item is available for sale.
- `is_delete` – Used for soft deletion (preserves data history).
- `created_at`, `updated_at` – Timestamps for record management.

Interpretation:

This table is central to stock control. It ensures that ordering and reporting modules access accurate inventory data. `is_active` and `is_delete` make it easier to manage visibility without permanently removing records.

Table: order_items

Purpose:

Represents the individual products included within each order. Each row links a specific `inventory` item to an order.

Key Columns:

- `id` (*PK*) – Unique identifier for the order line.
- `order_id` (*FK*) – References `orders.id`.
- `inventory_id` (*FK*) – References `inventory.id`.
- `qty` – Quantity of this item in the order.

- `unit_price` – Price per item at the time of sale.
- `line_total` – Total cost (`qty * unit_price`).
- **Interpretation:**

This is the **junction table** connecting orders to products. It enables multi-item orders and captures price/quantity at transaction time — preserving historical accuracy even if product prices later change.

 Table: orders

Purpose:

Stores all customer transactions — each record is one completed or pending order.

Key Columns:

- `id` (*PK*) – Unique order ID.
- `user_id` (*FK*) – References the users table (who processed the order).
- `customer_name` – Name of the customer (optional for walk-ins).
- `order_type` – e.g., “Dine-in” or “Take-out.”
- `takeout_fee` – Additional charge if applicable.
- `total_amount` – Overall total (sum of all `line_totals`).
- `amount_paid` – Cash received from the customer.
- `change_due` – Computed as `amount_paid - total_amount`.
- `status` – Current state (e.g., “pending,” “completed,” “cancelled”).
- `created_at` – Timestamp of order creation.

Interpretation:

This table forms the **transaction core** of the system. It ties together customers, items, and staff. The presence of status allows tracking of order progress and reporting on sales trends.

Table: users

Purpose:

Holds all system users, including admins and staff members who access the system.

Key Columns:

- `id` (*PK*) – Unique user ID.
- `username`, `password` – Login credentials.
- `full_name` – Display name of the user.
- `role` – Defines access rights (`admin`, `staff`).
- `created_at` – When the user account was created.

Interpretation:

This table supports authentication and authorization. Role-based access ensures separation of privileges — admins manage users and inventory, while staff handle daily ordering and sales.

Overall Database Interpretation

- The **users** table controls access.
- The **orders** table records each sale and links to the user who made it.
- The **order_items** table breaks each order into line items, connecting back to **inventory**.
- The **inventory** table maintains stock levels and product details.

Flow Summary:

users → orders → order_items → inventory

Together, these tables create a **fully normalized and trackable POS/ordering system**, supporting:

-  Inventory control
-  Order processing
-  Sales reporting
-  Role-based user management